PES University Hospital

AI Chatbot for OPD Patient Registration

Software Requirements Specification (SRS)

Project Name	AI Chatbot — OPD Patient Registration
Hospital	PES University Hospital, Bengaluru
Team	Student Internship Team — PES University
Tech Stack	MERN (MongoDB, Express, React, Node.js)
Version	1.0
Date	March 2026

1. What is this document?

This document explains what our chatbot does, why we are building it, how it works, and how we plan to build it. We have written it in simple language so anyone — whether technical or not — can read and understand it.

This is a student internship project done for PES University Hospital, Bengaluru.

2. What is the problem we are solving?

PES Hospital gets around 1,800 to 2,000 patients every single day at the OPD (Out-Patient Department). Every one of them has to go to the registration counter first before seeing a doctor.

Here is what happens at the counter right now:

Step 1	Patient walks in and stands in a queue at the counter
Step 2	Counter staff asks for their Aadhaar card (compulsory) and collects details — name, age, address, complaint
Step 3	Staff manually types everything into the hospital system
Step 4	Based on what the patient says, staff decides which OPD department to send them to
Step 5	Patient gets a token slip and goes to that OPD floor

This process has three real problems:

Problem	What actually happens because of it
Long queues	Patients wait 20–40 minutes just to register, especially in morning peak hours between 9 AM and 11 AM
Language barrier	Patients speak Kannada, Telugu, Tamil, Hindi and more. Counter staff cannot always understand everyone, leading to wrong department allocation
Manual data entry	Typing everything by hand is slow and sometimes creates duplicate patient records in the system

3. What is our solution?

We are building a chatbot — basically a smart registration form that talks to the patient in their own language. The patient can open it on their phone by scanning a QR code at the hospital entrance, or use a tablet that the hospital keeps at the entrance (called a kiosk).

The chatbot does the registration job **before** the patient reaches the counter. The counter still exists and still works — it just gets far fewer people standing in front of it.

No app download needed. It opens directly in the phone's browser, just like a website.

The counter remains open for:

- Elderly patients who are not comfortable with phones
- Patients without a smartphone
- Any complicated cases that need human help

4. How does the chatbot work? — Step by step

Here is exactly what happens when a patient uses our chatbot:

1. Patient arrives at the hospital entrance and either scans the QR code on their phone or walks up to the kiosk tablet.
2. The chatbot opens — no login, no account, no password needed.
3. Patient picks their language — Kannada, Hindi, English, Telugu, or Tamil. The entire chatbot switches to that language instantly.
4. Chatbot asks: are you a new patient or have you visited before?
5. If new patient — chatbot collects: name, age, gender, Aadhaar number, and mobile number. A unique Patient ID (called UHID) is automatically created and saved.
6. If returning patient — they type their existing UHID. Their details are already saved, so they skip the full form.
7. Patient taps big symptom buttons to describe what is wrong — for example: Fever, Eye problem, Chest pain, Stomach pain, Bone/Joint pain, Skin problem, Child health, Ear/Nose/Throat.
8. Chatbot looks at the selected symptoms and tells the patient which OPD department and floor to go to. For example: Fever → General Medicine, Floor 1, Room 101.
9. A token slip appears on the patient's phone screen showing their token number, department, floor, and time. They show this screen to the staff at the OPD entrance — exactly like showing a paper slip today.

Patient flow diagram

The diagram below shows the complete journey visually:

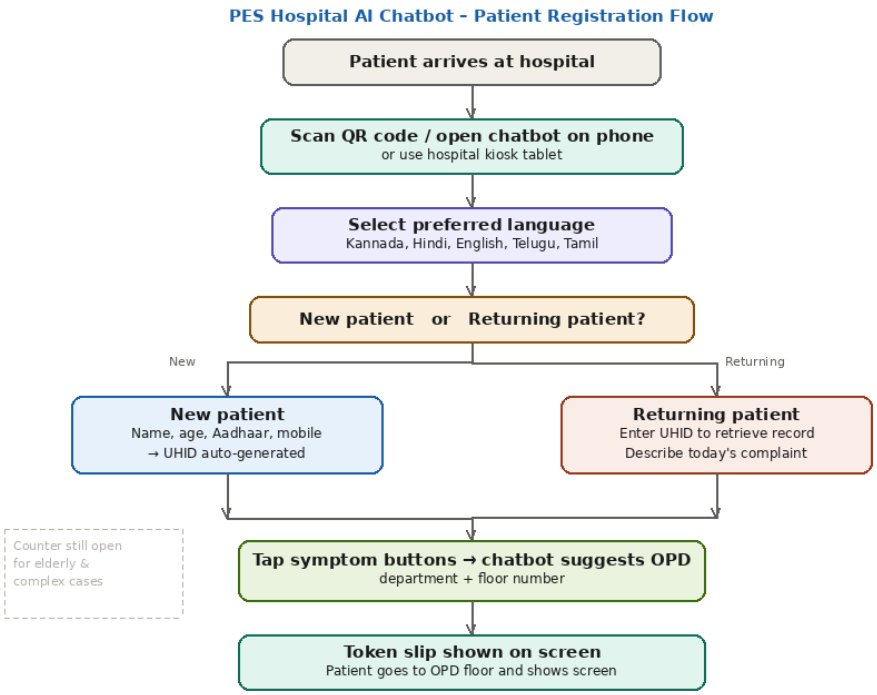

Figure 1 — How a patient moves through the chatbot from arrival to token

5. What exactly does the chatbot do?

5.1 For the patient

- Greets them in their own language
- Asks simple questions one at a time — like a conversation, not a big form
- Reads questions out loud using voice so even low-literacy patients can follow
- Accepts voice input — patient can speak their complaint instead of typing
- Shows big tap-friendly buttons for symptoms — no spelling, no typing
- Tells them exactly which floor and room to go to
- Shows a clean token slip on screen with all their details

5.2 For the counter staff

- A simple dashboard page shows all patients who registered through the chatbot today
- Staff can see each patient's name, token number, department, and time of registration
- Staff can still manually register patients who walked directly to the counter
- Both chatbot patients and counter patients appear in the same queue view

5.3 For the hospital admin

- A graphs page shows useful daily data — how many patients per department, what time the rush happens, which languages patients use most
- This is data the hospital currently does not have — our system generates it automatically as a side effect of digital registration

5.4 How does it decide which department to send the patient to?

We maintain a simple list that maps symptoms to departments. When the patient taps their symptoms, the system checks this list and gives the matching department. Examples:

Patient taps these symptoms	Goes to this department	Floor
Fever, cold, cough, body ache, weakness	**General Medicine**	Floor 1
Chest pain, heart racing, breathlessness	**Cardiology**	Floor 2
Eye pain, blurry vision, redness	**Ophthalmology**	Floor 2
Stomach pain, vomiting, diarrhoea, acidity	**Gastroenterology**	Floor 1
Bone pain, joint pain, back pain	**Orthopaedics**	Floor 3
Skin rash, itching, hair loss	**Dermatology**	Floor 3
Child fever, vaccination, child growth	**Paediatrics**	Floor 4
Toothache, gum pain, cavity	**Dental**	Floor 1
Ear pain, hearing problem, throat pain	**ENT**	Floor 2
Anything not matched above	**General Medicine (default)**	Floor 1

The hospital admin can update this list anytime without touching the code — it is stored in the database.

6. What we are NOT doing

To keep the project realistic for an 8-week internship, we have clearly decided NOT to build the following:

Not in our project	Why we left it out
Doctor appointment scheduling	That is a separate hospital system — too complex for this scope
OPD floor management after patient arrives	Once the patient has their token and goes to the floor, our job is done
Payment or billing	Completely separate system, not related to registration
Medical diagnosis	We only suggest a department — we are not doctors
Emergency handling	People in real emergencies run straight to the emergency ward — they do not stop to scan a QR code
Integration with existing hospital software	Their current system is separate — connecting to it is a much bigger project
Mobile app (Android/iOS)	Web app is enough — no download needed, works on any phone browser

7. What tools and technologies are we using?

Our team has decided to use the MERN stack. Here is what each part does in simple terms:

Name	What it is	What it does in our project
MongoDB	Database	Stores all patient information, visit records, and token numbers. Free cloud version (Atlas) used.
Express.js	Backend framework	Handles the logic on the server — saves patient data, generates UHID, generates token number, sends data back to the screen.
React.js	Frontend / UI	Builds the chatbot screens the patient sees — language selection, symptom buttons, token slip screen, counter dashboard.
Node.js	Server	Runs Express.js. Think of it as the engine that keeps the backend alive.
i18next	Translation library	Handles switching all text to the patient's chosen language. We write translation files for each language once.
Web Speech API	Voice (built into browser)	Lets the chatbot speak questions aloud and also accept voice input from the patient. Completely free — no API key needed.
Recharts	Graphs library for React	Draws the bar charts and graphs on the admin dashboard. Free and simple to use.
Vercel	Hosting for frontend	Puts our React app live on the internet. Free for students.
Render	Hosting for backend	Puts our Node/Express server live on the internet. Free tier available.

8. How is the data stored?

We use two separate storage buckets in MongoDB — one for patient details and one for each visit. This way a patient's personal information is saved only once, even if they visit the hospital 10 times.

Collection (bucket)	What it stores
patients	Name, age, gender, Aadhaar number, mobile, address, UHID. Saved once when patient registers for the first time.
visits	Which department, which symptoms selected, token number, date and time, new or follow-up visit. A new entry is created every single visit.

Token numbers are generated automatically. The system counts how many patients have already gone to a department today and adds 1. So the 47th person going to General Medicine today gets token GM-047. It resets to GM-001 every morning.

9. How are we going to build this?

We are building this in 5 stages over 8 weeks. The most important rule is: get the basic thing working first, then add extras on top. We do not try to build everything at once.

Stage	When	What we build
Stage 1	Week 1–2	React UI only — language selection screen, symptom picker buttons, token slip screen. No backend yet. Everything is hardcoded just to see how it looks and flows.
Stage 2	Week 3	Backend and database — Node + Express server, MongoDB connected, patient data saved, UHID auto-generated, token auto-generated, sent back to the screen.
Stage 3	Week 4	Connect everything — React calls the real backend. Full flow works end to end. Patient registers → data saved → token shown. This is the first real working demo.
Stage 4	Week 5–6	Voice and languages — Web Speech API added for voice input and output. i18next added for language switching. More symptom buttons added based on hospital feedback.
Stage 5	Week 7–8	Dashboard and graphs — Counter staff queue view, daily patient count graphs, department-wise charts, peak hour chart. Testing with real hospital staff. Final demo.

10. Who is doing what?

Frontend	Building the chatbot screens in React — what the patient sees and taps
Backend	Building the server in Node.js + Express — saving data, generating tokens, sending responses
Database	Setting up MongoDB Atlas, designing the patient and visit collections
Voice and Languages	Integrating Web Speech API for voice and i18next for multilingual support
Dashboard and Graphs	Building the counter staff view and admin analytics page using Recharts
Testing	Testing the full flow with real users at the hospital, fixing issues, preparing demo

11. One line summary of the whole project

Patient opens chatbot → picks language → taps symptoms → gets token and floor number → goes to OPD. Counter staff sees everything on a live dashboard with graphs.

That is it. Everything in this document is just explaining how we build that one thing.